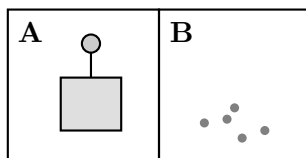


Intelligent Agents (AIMA 4th ed., Chapter 2)

Q1. The Vacuum World, Formalized

Consider the vacuum world of Figure 2.2, generalized to a $1 \times n$ row of squares ($n \geq 2$). Each square is either *Clean* or *Dirty*. The agent occupies exactly one square at a time. Its sensors report the label of its current square and whether that square contains dirt. Its actions are $\{Left, Right, Suck, NoOp\}$. Sucking cleans the current square, clean squares stay clean, and a move that would leave the row leaves the agent where it is.



- (a) State precisely the difference between an **agent function** and an **agent program**. Which of the two can depend on the entire percept history, and what must the other one do to obtain the same behavior?
- (b) How many distinct **environment states** are there? Justify your count.
- (c) How many distinct percepts can the agent receive? How many distinct percept sequences of length at most T are there? Give a closed form.

- (d) How many distinct agent functions are there over percept sequences of length at most T ? Evaluate your expression for $n = 2$, $T = 2$, and say in one sentence what the number implies about building agents by table lookup.

Q2. Rationality Is Relative

Work in the two-square world (squares A and B). Consider the agent whose agent function is: *if the current square is dirty then Suck; otherwise move to the other square*. Assume the geography is known a priori, but the initial dirt distribution and initial location are not; sensors are accurate; clean squares stay clean.

- (a) Under performance measure M_1 — one point for each clean square at each time step over a lifetime of 1000 steps — is this agent rational? Justify against the four things rationality depends on.

- (b) Now use M_2 : identical to M_1 , but with a penalty of one point for each movement action. Is the same agent still rational? Quantify the loss.

- (c) Show that *no* simple reflex agent is rational under M_2 , and name the smallest capability that has to be added.

- (d) Suppose clean squares can spontaneously become dirty again. Which environment properties change, and why is “*NoOp* forever” no longer rational?

- (e) Remove the location sensor: percepts are now only [*Dirty*] or [*Clean*]. Argue that no deterministic simple reflex agent cleans both squares from every start state, and show that a randomized one does. If the agent flips a fair coin between *Left* and *Right* on [*Clean*], what is the expected number of actions to reach the other square?

Q3. PEAS and the Seven Dimensions

A university deploys an agent on its inbound mail gateway. For each arriving message the agent chooses one action: *deliver*, *deliver with a warning banner*, *quarantine*, or *escalate to a human analyst*. Attackers adapt their campaigns in response to what gets through.

- (a) Give a PEAS description for this task environment. Be specific; “be accurate” is not a performance measure.

- (b) Classify this environment along all seven dimensions and justify the two you find most debatable.

- (c) Fill in the table. Use the vocabulary from AIMA Figure 2.6.

Task environment	Observable	Agents	Determ.	Episodic	Static	Discrete
Crossword puzzle						
Poker						
Warehouse pallet robot						
Mail-gateway agent (above)						

- (d) Why is **known vs. unknown** not a property of the environment at all? Give one environment that is known but partially observable, and one that is unknown but fully observable.

Q4. Choosing an Agent Architecture

For each scenario, select the **simplest** agent architecture that suffices. Assume each scenario is independent.

- (a) A thermostat switches the heater on whenever the measured temperature falls below the set-point and off when it rises above.
- ☐ simple reflex ☐ model-based reflex ☐ goal-based
☐ utility-based
- (b) A camera-based braking agent must decide whether the car ahead is braking. On older vehicles, taillights and brake lights are not distinguishable in a single frame — only the transition from off to on identifies braking.
- ☐ simple reflex ☐ model-based reflex ☐ goal-based
☐ utility-based
- (c) A campus delivery robot operates in a fully mapped building. Its destination changes from run to run, and it must plan a route to whichever room it is given.
- ☐ simple reflex ☐ model-based reflex ☐ goal-based
☐ utility-based
- (d) The same robot must now weigh delivery speed against battery consumption and the risk of collisions in crowded corridors, and no route satisfies all three.
- ☐ simple reflex ☐ model-based reflex ☐ goal-based
☐ utility-based
- (e) A vacuum agent with location and dirt sensors must stop moving once the entire floor is clean, under a performance measure that penalizes movement.
- ☐ simple reflex ☐ model-based reflex ☐ goal-based
☐ utility-based

Mark each statement **T** or **F** and give a one-line reason.

A simple reflex agent can be rational.	
Rationality requires knowing the actual outcome of one's actions.	
Every utility-based agent must also be a model-based agent.	
An autonomous agent is one that ignores knowledge supplied by its designer.	
In a fully observable, deterministic environment, an agent need not maintain internal state to keep track of the world.	
Randomized behavior is never rational.	

Q5. Learning Agents, Representations, and Bad Yardsticks

- (a) A learning agent has four conceptual components. Name the component responsible for each job below.

Selects the external action to take right now.	
Tells the agent how well it is doing against a fixed performance standard.	
Modifies the agent's components so that it does better in future.	
Suggests deliberately suboptimal actions that would produce informative experience.	

- (b) Why must the performance standard be *fixed* and conceptually *outside* the agent? What goes wrong otherwise?

- (c) Classify each representation as **atomic**, **factored**, or **structured**.

The state is the name of the city the driver is currently in.	
The state is a vector: GPS coordinates, fuel level, toll money, oil-light status.	
"A truck ahead is reversing into a driveway that a loose cow is blocking."	
The state is one of N hidden labels in a hidden Markov model.	
A complete assignment to the variables of a constraint satisfaction problem.	

- (d) Explain why a **distributed** representation degrades more gracefully under bit corruption than a **localist** one.

- (e) A vendor proposes to evaluate a cleaning agent by *the amount of dirt collected in an eight-hour shift*.
- (i) Describe the degenerate policy a rational agent would adopt under this measure.
 - (ii) Propose a measure that does not admit that exploit.
 - (iii) State the general design rule this illustrates, and say what an agent designer should do when the true measure cannot be written down correctly in advance.